Metropolis Random Walks Escape Deep Basins Faster in Higher Dimensions

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Abstract*—Simulated Annealing (SA) is a widely used stochastic optimization algorithm. It is of great interest to understand how its ability to escape local minima at a fixed temperature depends on geometry and dimension. In prior work, we derived closed-form expressions for the expected basin escape time of a constant-temperature SA process on a one-dimensional piecewise-linear landscape. In this paper, we extend the analysis to D-dimensional cubic basins defined by the infinity-norm energy $V(x) = \|x\|_\infty$. The walk proposes nearest-neighbor moves uniformly and escapes upon first reaching the outer shell, that is, any site with $\|x\|_\infty = N$. We construct a one-dimensional birth-death shell-quotient chain by collapsing each concentric shell in the Metropolis electrical network to a single node while retaining its total stationary mass. We derive a closed-form expression for the mean escape time $Q_D(N)$ of this shell-quotient chain as a function of the acceptance probability p, the basin depth N, and the dimension D. For fixed D and fixed $0 < p < 1$, the quotient escape time scales as $\Theta(\alpha^N/N^{D-1})$ as $N \to \infty$, where $\alpha = 1/p$. Relative to $D = 1$, this gives an asymptotic speedup of order N^{D-1} within this family of cubic basins. We prove that the true Metropolis mean escape time $H_D(N)$, without shell averaging, has the same $\Theta(\alpha^N/N^{D-1})$ scaling, establishing a lower bound via Rayleigh monotonicity and a matching upper bound through an explicit low-energy flow construction using Thomson's principle. Numerical solutions that exploit the walk's symmetries quantify the discrepancy between the quotient and true-chain means for $D = 1, 2, 3$.

I. Introduction

Simulated Annealing (SA), introduced by Kirkpatrick, Gelatt, and Vecchi [12], is a widely used stochastic optimization algorithm with applications ranging from Very Large Scale Integrated (VLSI) circuit placement [11] to quantum circuit compilation [10]. It mimics the physical process of bringing a system to its minimum energy state by controlling its temperature, using the Metropolis acceptance rule [13]. Prior works have applied convergence theory, spectral methods, and potential-theoretic metastability to studying similar escape dynamics, including in multidimensional settings [20], [21]. Here we seek an explicit discrete calculation that isolates the effect of dimension on the time to escape a basin.

In recent work [1], we developed a finite-time analytical framework for constant-temperature SA by studying a piecewise-linear energy landscape in one dimension. We modeled SA as a discrete-state Markov chain and derived closed-form expressions for the expected time to escape a single linear basin, both in the single-basin and two-basin settings. However, that analysis was restricted to one-dimensional landscapes. Real-world objective functions are typically high-dimensional, and a natural question is how the dimensionality of the landscape affects the escape dynamics. Note that a basin serves as an idealized model of trapping near a local minimum in an optimization landscape.

For the cubic basins studied here, higher fixed dimension reduces the asymptotic escape-time order as the basin depth grows. Although a D-dimensional cubic basin of radius N contains far more states than its one-dimensional counterpart, the number of escape routes across shells near the boundary grows as N^{D-1}, providing the polynomial factor in the escape-time scaling. The shell structure of the infinity-norm basin allows an exactly defined one-dimensional birth-death quotient chain to be constructed from its electrical network, making the quotient analysis tractable. For $D \geq 2$, the shell index of the true walk is generally not Markov. Nevertheless, shorting the shells yields a resistance estimate that gives an order-matching lower bound on the true walk's mean escape time, while an explicit flow construction gives a matching upper bound.

We study a Metropolis random walk on the integer lattice $\mathbb{Z}^D$ with energy $V(x) = \|x\|_\infty$. At each step, the walker proposes one of its $2D$ nearest-neighbor moves uniformly; downhill moves are always accepted, while uphill moves are accepted with probability $p = e^{-1/T}$. We derive exact shell-averaged proposal probabilities u_k and v_k using a binomial identity, and show that the resulting birth-death chain on the shell index admits a closed-form solution via a telescoping-product technique.

The key contributions of this work are:

1) We derive an exact closed-form expression for the mean escape time $Q_D(N)$ of the shell-quotient Metropolis chain on a D-dimensional cubic basin of depth N (Theorem 1).
2) We prove that the asymptotic scaling is $\Theta(\alpha^N/N^{D-1})$ for $0 < p < 1$ (Theorem 2), showing an asymptotic factor of order $1/N$ per additional fixed dimension, up to constants depending on dimension and p.
3) We establish that the shell-averaged chain is exactly the quotient chain obtained by shorting each shell in the true Metropolis network (Proposition 1), and use

Rayleigh monotonicity to prove a rigorous lower bound of $\Omega(\alpha^N/N^{D-1})$ for the true chain (Theorem 3).

4) We prove that the true Metropolis escape time has the same $\Theta(\alpha^N/N^{D-1})$ scaling (Theorem 4), by constructing an explicit low-energy flow to a face of the target shell and applying Thomson's principle to get an order-matching upper bound.
5) We compare the analytical results against numerical solutions of the finite-state Metropolis chain for $D \in \{1, 2, 3\}$.

The remainder of this paper is organized as follows. Section II reviews related work. Section III introduces the system model, including the D-dimensional cubic basin, the Metropolis random walk, and the shell-averaged proposal probabilities. Section IV derives the exact escape-time formula. Section V establishes the shell-quotient network construction, proves the asymptotic scaling, and proves the full $\Theta(\alpha^N/N^{D-1})$ law for the true chain. Section VI presents numerical results. Section VII concludes with a discussion of implications and future directions.

II. Related Work

The theoretical analysis of SA has progressed from asymptotic convergence guarantees to more practical finite-time characterizations. We organize the relevant literature into several threads.

Origins of simulated annealing. Kirkpatrick, Gelatt, and Vecchi [12] introduced simulated annealing as an optimization heuristic inspired by the Metropolis algorithm [13] from statistical mechanics. Geman and Geman [2] established global convergence under logarithmic cooling, and Hajek [3] clarified that schedules must effectively be slower than the escape time from the deepest local minimum. Catoni [4] developed rough large deviation estimates for simulated annealing with application to exponential schedules. These prior results establish that SA converges to the global optimum under appropriate cooling, but they do not provide tractable expressions for finite-time behavior at a fixed temperature.

Finite-time and spectral methods. Desai and Rao [5] studied the finite-time behavior of slowly cooled annealing chains. Holley, Kusuoka, and Stroock [6] analyzed asymptotics of the spectral gap with applications to the theory of simulated annealing. Mitra, Romeo, and Sangiovanni-Vincentelli [7] studied convergence and finite-time behavior of simulated annealing, establishing bounds on the probability of finding optimal solutions within a given number of steps. While these approaches yield rigorous general bounds, they often take the form of inequalities rather than exact expressions.

The one-dimensional predecessor. In [1], we adopted the notion of basin depth and focused on finite-time, constant-temperature regimes, quantifying the expected time to escape. We derived exact closed-form expressions for the mean escape time from a single linear basin in one dimension, and extended the analysis to a two-basin landscape to obtain exact expressions for the expected time to reach the global optimum. We also demonstrated how the predicted escape time can guide the design of a two-temperature switching strategy. The present paper extends that single-basin setting to cubic basins in any fixed dimension D: it gives an exact shell-quotient formula and a matching-order theorem for the true walk.

Potential-theoretic metastability. Bovier, Eckhoff, Gayrard, and Klein [20] obtain precise mean-exit-time and spectral results for a class of reversible discrete-state Markov chains under metastability assumptions. Bovier and den Hollander [21] develop the potential-theoretic framework, relating hitting probabilities and times to capacities and variational principles, with applications beyond one dimension. Within this potential-theoretic framework, our contribution is an explicit calculation for cubic basins in the growing-depth limit $N \to \infty$ at fixed D and p: an exact shell-quotient formula and a constructive matching-order resistance analysis that identifies the factor $N^{-(D-1)}$.

Exact and approximate state aggregation. Kemeny and Snell [22, Sec. 6.3, Thm. 6.3.2] show that grouping states preserves the Markov dynamics when all states in each group have the same total transition probability to every other group. Grouping all points in a shell generally fails this condition for $D \geq 2$, as illustrated in Section III. For the numerical computations in Section VI, we instead group points that differ only by coordinate signs or coordinate order. These points are equivalent by symmetry, so this smaller numerical model preserves the true walk's escape time exactly.

Random walks and first-passage times. Palyulin and Metzler [8] studied how a finite potential barrier affects the mean first-passage time, a problem closely related to our barrier-crossing framework. Roberts, Gelman, and Gilks [9] analyzed diffusion limits and optimal scaling of random walk Metropolis algorithms, providing connections between discrete Metropolis chains and continuous diffusion processes that motivate the discretization corrections studied in [1].

Geometry in high-dimensional optimization. The idea that high-dimensional geometry can accelerate escape from local minima has appeared in several contexts. Dauphin et al. [17] argue that saddle points and their surrounding plateaus, rather than high-error local minima, can be a major obstacle in high-dimensional non-convex optimization. In combinatorial optimization, Bellitti, Ricci-Tersenghi, and Scardicchio [18] demonstrated that for random constraint satisfaction problems, the dominant obstacle for both classical and quantum algorithms can be *entropic* rather than energetic. In biophysics, Bicout and Szabo [19] study diffusion in a spherically symmetric model of protein folding, illustrating entropic barriers and folding kinetics. These related geometric considerations motivate the present analysis.

Electrical network methods. The Rayleigh monotonicity principle and Thomson's variational principle for electrical networks provide powerful tools for bounding hitting times of reversible Markov chains [14]–[16]. We use these tools in Sections V-D and V-E to establish the full scaling law for the true Metropolis chain. Rayleigh monotonicity (shorting nodes can only decrease resistance) gives the lower bound, while Thomson's principle (any unit flow upper-bounds the

resistance) gives the matching upper bound via an explicit flow construction.

III. System Model

A. The D-Dimensional Cubic Basin

Consider a D-dimensional cubic basin on the integer lattice $\mathbb{Z}^D$ with energy function

$$V(x) = \|x\|_\infty = \max(|x_1|, \ldots, |x_D|). \tag{1}$$

The minimum-energy state is the origin, and the energy increases by one unit with each concentric shell. We define the *shell* at level k as

$$A_k := \{x \in \mathbb{Z}^D : \|x\|_\infty = k\}, \quad k = 0, 1, \ldots, N,$$

where N is the basin depth. Shell N is treated as absorbing: once the walker reaches any state with $\|x\|_\infty = N$, the escape is complete.

For a site on shell $k \geq 1$, let s denote the number of coordinates with $|x_i| = k$; we call these the *boundary coordinates*. The number of sites with exactly s boundary coordinates on shell k is

$$M_{D,s}(k) = \binom{D}{s} 2^s (2k-1)^{D-s}, \quad s = 1, \ldots, D. \tag{2}$$

There are $\binom{D}{s}$ ways to choose which s coordinates are at $\pm k$, 2^s sign choices for those coordinates, and $(2k-1)^{D-s}$ values in $\{-(k-1), \ldots, k-1\}$ for the remaining coordinates. The total number of sites on shell k is therefore

$$S_D(k) = \sum_{s=1}^{D} M_{D,s}(k) = (2k+1)^D - (2k-1)^D. \tag{3}$$

The identity (3) follows from the binomial theorem: $(2k+1)^D$ counts all integer points in $\{-k, \ldots, k\}^D$, and $(2k-1)^D$ counts those strictly inside. We use $S_D(0) := 1$ separately; formula (3) applies for $k \geq 1$.

Figure 1 illustrates the geometry for $D = 2$: the shells form concentric square terraces, with each outward step increasing the energy by one.

B. The Metropolis Random Walk

We study a fixed-temperature Metropolis random walk on $\mathbb{Z}^D$ with energy (1). At each time step, the walker at state x proposes one of the $2D$ nearest-neighbor moves $x \pm e_i$ uniformly at random, where e_i is the i-th standard basis vector. If the proposed move decreases the energy ($V(y) < V(x)$), it is always accepted. If the move increases the energy ($V(y) > V(x)$), it is accepted with probability

$$p = e^{-\Delta E/T} = e^{-1/T}, \tag{4}$$

since each uphill step increases V by exactly one unit. Moves that keep the energy unchanged are accepted with probability 1.

We write $\alpha := 1/p \geq 1$ for the *bias ratio*; $\alpha > 1$ when $p < 1$, governing the exponential barrier to escape. The first-passage time is

$$\tau_N := \inf\{t \geq 0 : \|X_t\|_\infty = N\}, \tag{5}$$

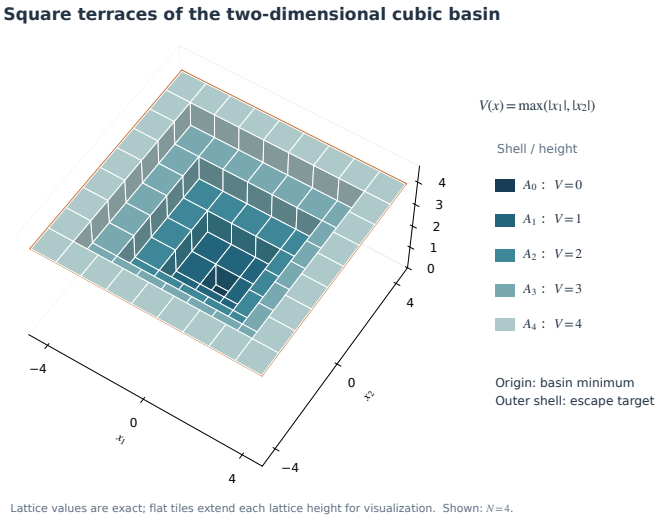


Fig. 1. The cubic basin for $D = 2$ and $N = 4$. Each square terrace represents a shell A_k at energy $V(x) = \|x\|_\infty = k$. The origin is the basin minimum; escape occurs upon reaching the outermost shell A_N. The flat tiles extend the discrete lattice heights to visualize the stepped geometry.

starting from the origin $X_0 = 0$. We distinguish the true-chain mean escape time $H_D(N) := \mathbb{E}_0[\tau_N]$ from the shell-quotient mean escape time $Q_D(N)$ defined below. The dependence on p is suppressed in both notations.

C. Shell-Averaged Proposal Probabilities

We define a simplified chain on the shell indices by averaging the outward and inward proposal probabilities uniformly over all sites in each shell. For $D \geq 2$, these probabilities vary within a shell, so the resulting shell-quotient chain does not exactly describe the true walk's shell transitions. For example, at $(k, 0, \ldots, 0)$ only one proposal direction leads outward, whereas at $(k, k, 0, \ldots, 0)$ two do, giving outward proposal probabilities $1/(2D)$ and $1/D$, respectively, for $1 \leq k < N$.

A site with s boundary coordinates has s outward directions (each increasing $\|x\|_\infty$ from k to $k+1$) out of $2D$ total directions. The outward proposal probability averaged over shell k is

$$u_k = \frac{1}{2D\, S_D(k)} \sum_{s=1}^{D} s\, M_{D,s}(k). \tag{6}$$

Substituting (2) gives $u_k = \frac{1}{2D\, S_D(k)} \sum_{s=1}^{D} s\binom{D}{s} 2^s (2k-1)^{D-s}$. We simplify this using the binomial identity

$$\sum_{s=1}^{D} s \binom{D}{s} 2^s a^{D-s} = 2D(a+2)^{D-1}, \tag{7}$$

which can be derived by differentiating $(a+2)^D = \sum_{s=0}^{D} \binom{D}{s} 2^s a^{D-s}$ with respect to a, multiplying by a, and subtracting from $D(a+2)^D$. Setting $a = 2k-1$ yields $\sum_{s=1}^{D} s\, M_{D,s}(k) = 2D(2k+1)^{D-1}$, and therefore

$$\boxed{u_k = \frac{(2k+1)^{D-1}}{S_D(k)}.} \tag{8}$$

For the inward direction, only sites with exactly $s = 1$ boundary coordinate can step inward: if $s \geq 2$, reducing the absolute value of one boundary coordinate from k to $k-1$ leaves $\|x\|_\infty = k$ unchanged. Each such site has one inward direction out of $2D$, and the number of such sites is $M_{D,1}(k) = 2D(2k-1)^{D-1}$. Therefore

$$\boxed{v_k = \frac{(2k-1)^{D-1}}{S_D(k)}.} \tag{9}$$

Note that u_k and v_k are *proposal* probabilities. Since outward moves are accepted with probability p and inward moves are always accepted, the actual shell-averaged transition probabilities are $p\,u_k$ (outward) and v_k (inward). The remaining probability $r_k = 1 - p\,u_k - v_k$ accounts for same-shell moves (both accepted lateral proposals and rejected uphill proposals).

Intuitively, for large k, most sites on a shell have exactly one boundary coordinate, giving one outward and one inward proposal direction. Accordingly, both u_k and v_k approach $1/(2D)$ as $k \to \infty$. The corresponding outward transition probability approaches $p/(2D)$.

IV. Exact Escape Time

A. Birth-Death Recurrence

Let T_k denote the mean first-passage time from state k to the absorbing state N of the shell-quotient birth-death chain. We retain T_k in this derivation and write $Q_D(N) := T_0^{(D)}(N)$ for its escape time from state 0. All recurrences below with $k \geq 1$ apply to transient indices $1 \leq k < N$, with boundary condition $T_N = 0$. The first-step equation for $k \geq 1$ is

$$T_k = 1 + p\,u_k\,T_{k+1} + v_k\,T_{k-1} + r_k\,T_k.$$

Substituting $r_k = 1 - p\,u_k - v_k$ and rearranging, we obtain

$$p\,u_k\,(T_k - T_{k+1}) - v_k\,(T_{k-1} - T_k) = 1. \tag{10}$$

We define the *backward differences* $b_k := T_k - T_{k+1}$, so that (10) becomes

$$\boxed{p\,u_k\,b_k = v_k\,b_{k-1} + 1, \qquad k \geq 1.} \tag{11}$$

At the origin ($k = 0$), all $2D$ neighbors lie on shell 1, and each proposed move is uphill with acceptance probability p. By symmetry, $T_0 = 1 + p\,T_1 + (1-p)\,T_0$, which gives

$$\boxed{b_0 := T_0 - T_1 = \frac{1}{p}.} \tag{12}$$

We define the ratio

$$R_k := \frac{v_k}{p\,u_k} = \frac{1}{p}\left(\frac{2k-1}{2k+1}\right)^{D-1}, \tag{13}$$

so the recurrence can be written as $b_k = R_k\,b_{k-1} + 1/(p\,u_k)$.

Unrolling the first few terms:

$$b_1 = R_1\,b_0 + \frac{1}{p\,u_1},$$

$$b_2 = R_2R_1\,b_0 + R_2\frac{1}{p\,u_1} + \frac{1}{p\,u_2},$$

$$b_3 = R_3R_2R_1\,b_0 + R_3R_2\frac{1}{p\,u_1} + R_3\frac{1}{p\,u_2} + \frac{1}{p\,u_3}.$$

This suggests, and an induction argument confirms, the general form

$$b_k = \left(\prod_{j=1}^{k} R_j\right) b_0 + \sum_{m=1}^{k} \frac{1}{p\,u_m} \prod_{j=m+1}^{k} R_j. \tag{14}$$

B. Telescoping and Closed Form

The product of ratios telescopes. From (13), for $k \geq m+1$,

$$\prod_{j=m+1}^{k} R_j = \frac{1}{p^{k-m}} \prod_{j=m+1}^{k} \left(\frac{2j-1}{2j+1}\right)^{D-1}.$$

The inner product telescopes:

$$\prod_{j=m+1}^{k} \frac{2j-1}{2j+1} = \frac{2m+1}{2m+3} \cdot \frac{2m+3}{2m+5} \cdots \frac{2k-1}{2k+1} = \frac{2m+1}{2k+1}.$$

Therefore

$$\prod_{j=m+1}^{k} R_j = \frac{1}{p^{k-m}} \left(\frac{2m+1}{2k+1}\right)^{D-1}. \tag{15}$$

Substituting (15) into (14) with $b_0 = 1/p$ and $1/(p\,u_m) = S_D(m)/\big(p(2m+1)^{D-1}\big)$:

The $m = 0$ term (from b_0):

$$\left(\prod_{j=1}^{k} R_j\right) \frac{1}{p} = \frac{1}{p^{k+1}(2k+1)^{D-1}}.$$

The $m \geq 1$ terms: the factors $(2m+1)^{D-1}$ cancel between $1/(p\,u_m)$ and the telescoping product, giving

$$\frac{1}{p\,u_m} \prod_{j=m+1}^{k} R_j = \frac{S_D(m)}{p^{k-m+1}(2k+1)^{D-1}}.$$

Combining and writing $\alpha = 1/p$, we obtain

$$\boxed{b_k = \frac{1}{(2k+1)^{D-1}} \left[\alpha^{k+1} + \sum_{m=1}^{k} S_D(m)\,\alpha^{k-m+1}\right].} \tag{16}$$

Since $T_N = 0$ and $b_\ell = T_\ell - T_{\ell+1}$, we have $T_0 = \sum_{\ell=0}^{N-1} b_\ell$. This yields our first main result.

Theorem 1 (Exact escape time)**.** *For the shell-averaged Metropolis random walk on a D-dimensional cubic basin of depth N with acceptance probability $p \in (0,1]$ and $\alpha = 1/p$, the mean escape time from the origin is*

$$\boxed{T_0^{(D)}(N) = \sum_{\ell=0}^{N-1} \frac{1}{(2\ell+1)^{D-1}} \left[\alpha^{\ell+1} + \sum_{m=1}^{\ell} S_D(m)\,\alpha^{\ell-m+1}\right],} \tag{17}$$

where $S_D(m) = (2m+1)^D - (2m-1)^D$ is the number of lattice points on shell m. Thus this formula gives $Q_D(N)$ exactly. It also gives $H_1(N)$ when $D = 1$, but is used as an approximation to $H_D(N)$ for $D \geq 2$.

Each term b_ℓ in the sum can be interpreted as follows: the factor $\alpha^{\ell+1}$ captures the exponential cost of climbing from shell ℓ to shell $\ell+1$, the shell-size terms $S_D(m)$ weight the contributions of intermediate shells, and the prefactor $1/(2\ell+1)^{D-1}$ captures the speedup with higher dimensions: the larger the shell surface, the easier the outward step.

C. Special Cases

We verify Theorem 1 by examining low-dimensional special cases.

$D = 1$: Here $S_1(m) = 2$ and $(2\ell+1)^{D-1} = 1$, so

$$T_0^{(1)} = \sum_{\ell=0}^{N-1} \left[\alpha^{\ell+1} + 2 \sum_{m=1}^{\ell} \alpha^{\ell-m+1} \right]. \tag{18}$$

Remark 1 (Recovery of the one-dimensional result)**.** *Formula* (18) *coincides with the single-basin escape time derived in [1], after identifying our acceptance probability p with $2p_{\text{prior}}$, where the factor of two accounts for the fact that [1] absorbs the one-dimensional proposal probability $1/2$ into the definition of p. In both cases, $\alpha = e^{1/T}$ represents the same physical bias ratio.*

$D = 2$: Here $S_2(m) = (2m+1)^2 - (2m-1)^2 = 8m$, and the formula becomes

$$T_0^{(2)} = \sum_{\ell=0}^{N-1} \frac{1}{2\ell+1} \left[\alpha^{\ell+1} + \sum_{m=1}^{\ell} 8m\, \alpha^{\ell-m+1} \right]. \tag{19}$$

The $1/(2\ell+1)$ prefactor produces the $1/N$ asymptotic speedup relative to one dimension.

$D = 3$: Here $S_3(m) = (2m+1)^3 - (2m-1)^3 = 24m^2 + 2$, giving

$$T_0^{(3)} = \sum_{\ell=0}^{N-1} \frac{1}{(2\ell+1)^2} \left[\alpha^{\ell+1} + \sum_{m=1}^{\ell} (24m^2+2)\, \alpha^{\ell-m+1} \right]. \tag{20}$$

Remark 2 (The unbiased case $p = 1$)**.** *When $p = 1$ (so $\alpha = 1$), the Cartesian random walk is unbiased. The shell sizes telescope: $1 + \sum_{m=1}^{\ell} S_D(m) = (2\ell+1)^D$, so $b_\ell = (2\ell+1)^D/(2\ell+1)^{D-1} = 2\ell+1$. Thus $Q_D(N) = \sum_{\ell=0}^{N-1}(2\ell+1) = N^2$ for every D when $p = 1$. This dimension-independent identity applies to the quotient chain; the true-chain mean $H_D(N)$ need not equal N^2.*

V. Shell-Quotient Construction and Asymptotic Scaling

A. Construction of the Birth-Death Quotient

The shell-averaged chain is obtained by collapsing each shell in the true Metropolis electrical network to a single node, as illustrated in Figure 2. The node retains the shell's total stationary mass, while conductances between successive shells are added. Transitions within a shell become self-loops.

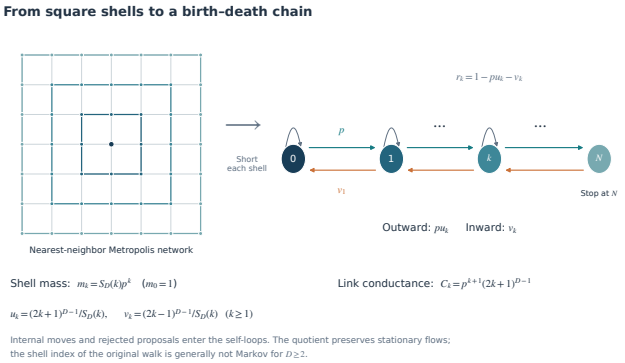


Fig. 2. Shell-quotient construction, illustrated for $D = 2$. Shorting each shell produces a birth–death chain with shell mass $m_k = S_D(k)p^k$ $(m_0 = 1)$ and link conductance $C_k = p^{k+1}(2k+1)^{D-1}$. For $1 \leq k < N$, outward and inward transitions have probabilities pu_k and v_k; internal moves and rejected proposals contribute to self-loops. Escape calculations stop at the first visit to N.

Proposition 1 (The shell-averaged chain is the quotient chain)**.** *Fix N and consider the true Metropolis chain restricted to $B_N := \{x \in \mathbb{Z}^D : \|x\|_\infty \leq N\}$, with any proposal leaving B_N rejected. This un-killed, reflecting chain is reversible with respect to*

$$\pi(x) \propto p^{\|x\|_\infty}. \tag{21}$$

Short each shell A_k to a single node and retain its total stationary mass. For $1 \leq k < N$, the quotient birth-death chain has transition probabilities $p\,u_k$ (outward) and v_k (inward); at $k = 0$ its outward probability is p. For escape-time calculations, stop either chain at its first visit to shell N.

Proof. Reversibility follows from detailed balance. If $x \sim y$ and $V(y) = V(x) + 1$, then

$$\pi(x)\, P(x,y) \propto p^{V(x)} \cdot \frac{p}{2D} = \frac{p^{V(y)}}{2D} \propto \pi(y)\, P(y,x),$$

since $P(y,x) = 1/(2D)$ for downhill moves. Equal-energy edges have symmetric proposal probabilities and equal weights, and rejected proposals contribute only self-loops. For the network calculation we use the unnormalized measure $\mu(x) = p^{\|x\|_\infty}$, so conductances below are exactly $c(x,y) = \mu(x)P(x,y)$, without a normalization factor.

The total stationary mass of shell A_k is proportional to $S_D(k)\,p^k$. In this normalization the shell mass is $S_D(k)p^k$, with $S_D(0) = 1$. We count the edges crossing from shell k to shell $k+1$. From the shell-counting identity (7), the total number of oriented outward edges is $2D(2k+1)^{D-1}$. Each such edge carries conductance $p^{k+1}/(2D)$ (from $\pi(x)\,P(x,y)$ with $V(x) = k$), so the total conductance across the cut $A_k \leftrightarrow A_{k+1}$ is

$$C_k = p^{k+1}(2k+1)^{D-1}. \tag{22}$$

The quotient chain therefore has outward transition probability

$$\widetilde{P}(k,k+1) = \frac{C_k}{S_D(k)\,p^k} = p\,\frac{(2k+1)^{D-1}}{S_D(k)} = p\,u_k,$$

and inward transition probability

$$\widetilde{P}(k,k-1) = \frac{C_{k-1}}{S_D(k)\,p^k} = \frac{p^k(2k-1)^{D-1}}{S_D(k)\,p^k} = \frac{(2k-1)^{D-1}}{S_D(k)} = v_k.$$

These are exactly the shell-averaged probabilities, completing the proof. The displayed inward and outward formulas apply for $1 \le k < N$. At $k = 0$, the $2D$ edges to A_1 each have conductance $p/(2D)$, so $C_0 = p$ and $\widetilde{P}(0,1) = p$. Internal transitions and rejected proposals contribute to the quotient self-loop, preserving the shell mass. □

B. Asymptotic Scaling

Theorem 2 (Asymptotic scaling)**.** *Fix $D \ge 1$ and $0 < p < 1$, and let $\alpha = 1/p > 1$. Then the shell-averaged escape time satisfies $Q_D(N) = T_0^{(D)}(N)$ and*

$$T_0^{(D)}(N) = \Theta\left(\frac{\alpha^N}{N^{D-1}}\right) \qquad (N \to \infty). \tag{23}$$

In particular, $T_0^{(1)}(N) = \Theta(\alpha^N)$, $T_0^{(2)}(N) = \Theta(\alpha^N/N)$, and $T_0^{(3)}(N) = \Theta(\alpha^N/N^2)$.

Proof. From (16), factor out $\alpha^{\ell+1}$:

$$b_\ell = \frac{\alpha^{\ell+1}}{(2\ell+1)^{D-1}}\,F_\ell, \tag{24}$$

where $F_\ell := 1 + \sum_{m=1}^{\ell} S_D(m)\,\alpha^{-m}$. Since $S_D(m) = \Theta(m^{D-1})$ and $\alpha > 1$, the series $\sum_{m\ge1} S_D(m)\,\alpha^{-m}$ converges. Hence F_ℓ is increasing and bounded: $1 \le F_\ell \le F_\infty < \infty$, where $F_\infty = 1 + \sum_{m\ge1} S_D(m)\,\alpha^{-m}$. It follows that $b_\ell \asymp \alpha^\ell/\ell^{D-1}$.

Next, the ratio of consecutive terms satisfies

$$\frac{b_{\ell-1}}{b_\ell} = \alpha^{-1}\left(\frac{2\ell+1}{2\ell-1}\right)^{D-1}\frac{F_{\ell-1}}{F_\ell} \longrightarrow \alpha^{-1} < 1.$$

Hence there exist $q \in (0,1)$ and L such that $b_{\ell-1} \le q\,b_\ell$ for all $\ell \ge L$. The sum $T_0^{(D)}(N) = \sum_{\ell=0}^{N-1} b_\ell$ is therefore geometrically dominated by its last term:

$$b_{N-1} \le T_0^{(D)}(N) \le \sum_{\ell=0}^{L-1} b_\ell + \frac{1}{1-q}\,b_{N-1} \le C'\,b_{N-1}.$$

Since $b_{N-1} = \Theta(\alpha^N/N^{D-1})$, the claim follows. □

Remark 3 (Sharper asymptotic)**.** *The proof also yields the sharper estimate*

$$T_0^{(D)}(N) \sim \frac{\alpha}{\alpha-1}\,F_\infty\,\frac{\alpha^N}{(2N)^{D-1}}. \tag{25}$$

To justify the leading constant, put $r := \alpha^{-1} < 1$. For each fixed $j \ge 0$, the ratio limit in the proof gives $b_{N-1-j}/b_{N-1} \to r^j$ as $N \to \infty$. Choose $r < q < 1$ and L such that $b_{\ell-1} \le qb_\ell$ for all $\ell \ge L$. The terms with $\ell \ge L$, read backward from $N-1$ and divided by b_{N-1}, are bounded by the summable sequence q^j. The fixed prefix with $\ell < L$ contributes a quantity tending to zero, since $b_{N-1} \to \infty$. Dominated convergence therefore gives

$$\frac{T_0^{(D)}(N)}{b_{N-1}} \longrightarrow \sum_{j=0}^{\infty} r^j = \frac{\alpha}{\alpha-1}.$$

Combining this with $b_{N-1} = \alpha^N F_{N-1}/(2N-1)^{D-1}$ and $F_{N-1} \to F_\infty$ proves the stated equivalent.

The scaling $\Theta(\alpha^N/N^{D-1})$ shows that escape time grows exponentially with N, with the base α determined by temperature. Higher dimension reduces this time by a polynomial factor because the outermost shells provide $\Theta(N^{D-1})$ escape routes.

C. Hitting-Time Identity

Throughout the resistance and hitting-time arguments, $D \ge 1$ and $0 < p < 1$ are fixed. Unless a finite restriction is stated explicitly, let $P(x,y)$ denote the one-step transition probability from x to y for the unkilled Metropolis chain on $\mathbb{Z}^D$, stopped only when its specified target is reached. Constants may depend on D and p, but not on N.

To relate escape time to effective resistance, we use the reversible measure $\mu(x) := p^{\|x\|_\infty}$, under which the chain satisfies detailed balance: $\mu(x)\,P(x,y) = \mu(y)\,P(y,x)$ for all adjacent x, y. The associated edge conductances are $c(x,y) = \mu(x)\,P(x,y)$, and the Dirichlet form is

$$\mathcal{E}(f,f) := \tfrac{1}{2}\sum_{x\sim y} c(x,y)\big(f(x)-f(y)\big)^2.$$

Here $\sum_{x\sim y}$ counts both orientations of each distinct nearest-neighbor edge. Equivalently, the Dirichlet form sums once over unoriented edges without the factor $1/2$. All flow-energy sums below count each unoriented edge once, using any fixed orientation and an antisymmetric flow $\theta(y,x) = -\theta(x,y)$:

$$\operatorname{div}\theta(x) := \sum_{y\sim x}\theta(x,y), \qquad \mathcal{E}_{\text{flow}}(\theta) := \sum_{\{x,y\}}\frac{\theta(x,y)^2}{c(x,y)}.$$

Self-loops do not contribute to either energy; their contribution to the holding times is retained through μ and P. The capacity between a point a and a set A is $\operatorname{Cap}(a,A) := \inf\{\mathcal{E}(f,f) : f(a) = 1, f|_A = 0\}$. By Dirichlet's principle [14], this minimum is the effective conductance between a and A. Hence $R_{\text{eff}}(a \leftrightarrow A) = 1/\operatorname{Cap}(a,A)$.

Lemma 1 (Hitting time via capacity)**.** *For a nonempty finite target $A \subset \mathbb{Z}^D$ and $a \notin A$, let $\tau_A := \inf\{t \ge 0 : X_t \in A\}$, $\tau_a := \tau_{\{a\}}$, and $\tau_a^+ := \inf\{t \ge 1 : X_t = a\}$. Then*

$$\mathbb{E}_a[\tau_A] = \frac{1}{\operatorname{Cap}(a,A)}\sum_x \mu(x)\Pr_x(\tau_a < \tau_A). \tag{26}$$

In particular,

$$\mu(a)\,R_{\text{eff}}(a \leftrightarrow A) \le \mathbb{E}_a[\tau_A] \le M_{D,p}\,R_{\text{eff}}(a \leftrightarrow A), \tag{27}$$

where $M_{D,p} := \sum_{x\in\mathbb{Z}^D}\mu(x) < \infty$.

Proof. The reversible measure $\mu(x) = p^{\|x\|_\infty}$ has finite total mass

$$M_{D,p} = 1 + \sum_{k\geq 1} S_D(k)p^k < \infty,$$

because $S_D(k) = O(k^{D-1})$ and $p < 1$. Thus the irreducible chain has stationary distribution $\mu/M_{D,p}$ and is positive recurrent. Consequently, the hitting and return times considered below have finite means.

We first establish the capacity identity. Put $h(x) = \Pr_x(\tau_a < \tau_A)$, the probability of reaching a before A. This function equals 1 at a and 0 on A. At every other state, conditioning on the first step gives

$$h(x) = \sum_y P(x,y)h(y).$$

For two functions, the corresponding mixed term is

$$\mathcal{E}(f,g) = \frac{1}{2}\sum_{x\sim y} c(x,y)\big(f(x)-f(y)\big)\big(g(x)-g(y)\big).$$

Expanding this sum and using reversibility, $c(x,y) = c(y,x)$, yields

$$\mathcal{E}(f,g) = \sum_x \mu(x)g(x)\left(f(x) - \sum_y P(x,y)f(y)\right).$$

Self-loops contribute zero to the energy and may be included when using $\sum_y P(x,y) = 1$.

Taking $f = g = h$, only the term at a remains: on A, h is zero, and elsewhere the expression in parentheses is zero. Hence

$$\begin{aligned}\mathcal{E}(h,h) &= \mu(a)\left(1 - \sum_y P(a,y)h(y)\right)\\ &= \mu(a)\Pr_a(\tau_A < \tau_a^+).\end{aligned}$$

The last equality follows by conditioning on the first step: $\sum_y P(a,y)h(y)$ is the probability of returning to a before reaching A.

We now show that h minimizes energy among functions equal to 1 at a and 0 on A. It suffices to consider functions with values in $[0,1]$, since truncating to this interval preserves the boundary values and cannot increase energy. For any such function f, write $f = h + g$. Then $g = 0$ on $\{a\}\cup A$, so

$$\mathcal{E}(h,g) = \sum_x \mu(x)g(x)\left(h(x) - \sum_y P(x,y)h(y)\right) = 0.$$

Indeed, g vanishes on the boundary, and the expression in parentheses vanishes elsewhere. Expanding the energy gives

$$\mathcal{E}(f,f) = \mathcal{E}(h,h) + 2\mathcal{E}(h,g) + \mathcal{E}(g,g) \geq \mathcal{E}(h,h).$$

All these sums converge absolutely because the functions are bounded and $\sum_x \mu(x) < \infty$. Thus h attains the infimum defining capacity, and

$$\mathrm{Cap}(a,A) = \mu(a)\Pr_a(\tau_A < \tau_a^+).$$

To obtain the expected hitting time, let

$$G_A(u,x) := \mathbb{E}_u\left[\sum_{t=0}^{\tau_A - 1} \mathbf{1}\{X_t = x\}\right]$$

be the expected number of visits to x before reaching A, starting from u. Summing over visited states gives

$$\mathbb{E}_a[\tau_A] = \sum_x G_A(a,x).$$

Reversibility, applied to paths that avoid A, gives

$$\mu(a)G_A(a,x) = \mu(x)G_A(x,a).$$

Starting from x, visits to a occur only if the chain reaches a before A. The strong Markov property therefore gives

$$G_A(x,a) = h(x)G_A(a,a).$$

Starting from a, each visit to a is followed by probability $q = \Pr_a(\tau_A < \tau_a^+)$ of reaching A before the next return. Thus the number of visits to a, including the initial visit, is geometric with mean $1/q$, so

$$G_A(a,a) = \frac{1}{q} = \frac{\mu(a)}{\mathrm{Cap}(a,A)}.$$

Combining these identities yields

$$\begin{aligned}\mathbb{E}_a[\tau_A] &= \frac{\sum_x \mu(x)h(x)}{\mathrm{Cap}(a,A)}\\ &= R_{\mathrm{eff}}(a\leftrightarrow A)\sum_x \mu(x)h(x),\end{aligned}$$

which is (26). Finally, $h(a) = 1$ and $0 \leq h(x) \leq 1$ imply

$$\mu(a) \leq \sum_x \mu(x)h(x) \leq M_{D,p},$$

giving (27). □

For the fixed starting point $a = 0$, where $\mu(0) = 1$, and fixed D, p, Lemma 1 compares the mean hitting time with effective resistance using constants independent of the target depth N. This enables both lower and upper bounds through resistance estimates.

D. Lower Bound for the True Chain

Figure 3 illustrates the two network constructions used to bound the true escape time. For the lower bound, shorting each shell decreases effective resistance and produces a series network with link resistances $1/C_k$, as shown in Figure 3(a).

Theorem 3 (Lower bound)**.** *Fix $D \geq 1$ and $0 < p < 1$, and let $\alpha = 1/p > 1$. For the true Metropolis chain started from the origin,*

$$\mathbb{E}_0[\tau_N] \geq c_{D,p}\,\frac{\alpha^N}{N^{D-1}} \qquad (N \geq 1), \tag{28}$$

for some constant $c_{D,p} > 0$ independent of N.

Proof. Shorting all vertices within each shell to a single node produces the quotient network of Proposition 1. Because A_N separates the origin from the exterior of B_N, the exterior

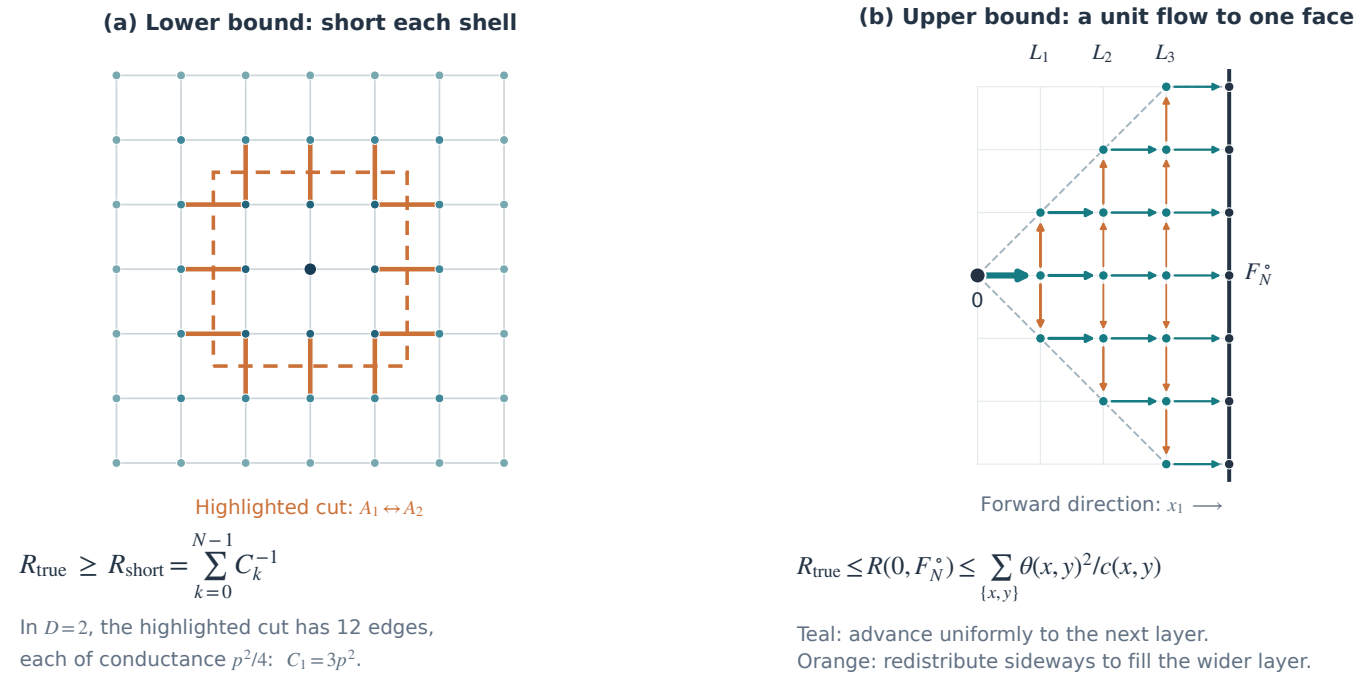


Fig. 3. Electrical-network bounds, illustrated for $D = 2$. (a) Shorting each shell gives the resistance lower bound $R_{\text{eff}}(0, A_N) \geq \sum_{k=0}^{N-1} C_k^{-1}$. The highlighted cut between A_1 and A_2 contains 12 edges, each of conductance $p^2/4$, giving $C_1 = 3p^2$. (b) A unit flow through an expanding pyramid subnetwork gives the resistance upper bound by Thomson's principle. Teal arrows carry current forward between layers; orange arrows redistribute it within each wider layer. The final step reaches the target face F_N°. The forward coordinate x_1 is drawn horizontally. Both constructions yield resistance bounds of order α^N/N^{D-1}.

does not affect the resistance to A_N: an equilibrium potential is zero there. We may therefore perform this shorting in B_N. By Rayleigh monotonicity [14], [15], shorting can only decrease resistance: $R_{\text{eff}}(0, A_N) \geq R_{\text{eff}}^{\text{coll}}(0, N)$. In the collapsed network, the conductance across cut $k \leftrightarrow k+1$ is $C_k = p^{k+1}(2k+1)^{D-1}$ from (22), so

$$R_{\text{eff}}^{\text{coll}}(0, N) = \sum_{k=0}^{N-1} \frac{1}{C_k} = \sum_{k=0}^{N-1} \frac{\alpha^{k+1}}{(2k+1)^{D-1}} = \Theta\left(\frac{\alpha^N}{N^{D-1}}\right).$$

The lower bound in (27) then gives $\mathbb{E}_0[\tau_N] \geq \mu(0)\, R_{\text{eff}}(0, A_N) \geq c_{D,p}\, \alpha^N/N^{D-1}$. In fact, retaining only the final term of the resistance sum gives

$$\mathbb{E}_0[\tau_N] \geq \frac{\alpha^N}{(2N-1)^{D-1}} \geq 2^{-(D-1)} \frac{\alpha^N}{N^{D-1}},$$

so one may take $c_{D,p} = 2^{-(D-1)}$ for all $N \geq 1$. □

Remark 4. *Rayleigh monotonicity compares effective resistances. Combined with Lemma 1, it gives the lower bound of order α^N/N^{D-1} on $H_D(N)$ in Theorem 3.*

E. Upper Bound via an Explicit Flow

We construct a unit current from the origin to a face of the target shell. Thomson's principle then bounds the effective resistance by the energy of this current. For a current θ on edges with conductances $c(e)$, its energy is

$$\mathcal{E}(\theta) = \sum_e \frac{\theta(e)^2}{c(e)},$$

where each unoriented edge is counted once. The construction spreads the current across progressively larger layers, reducing the energy needed to reach the target.

A large target face. Consider the face

$$F_N^\circ := \{(N, y_2, \ldots, y_D) : |y_i| \leq N-1,\ i = 2, \ldots, D\}$$

of the target shell A_N. It contains

$$|F_N^\circ| = (2N-1)^{D-1} = \Theta(N^{D-1})$$

vertices. Since every trajectory that reaches this face has also reached the shell,

$$\mathbb{E}_0[\tau_N] \leq \mathbb{E}_0[\tau_{F_N^\circ}].$$

It therefore suffices to upper-bound the expected time to reach F_N°.

A pyramid subnetwork. Write $n := D-1$ and let

$$Q_k := \{-k,\dots,k\}^n, \qquad |Q_k| = (2k+1)^n.$$

We use the expanding layers

$$L_k := \{(k,y) : y \in Q_k\}, \qquad k = 0,\dots,N-1,$$

followed by the target layer

$$L_N^\circ := \{(N,y) : y \in Q_{N-1}\} = F_N^\circ.$$

Within L_k, lateral edges have conductance $p^k/(2D)$. The forward edges from (k,y) to $(k+1,y)$ have conductance $p^{k+1}/(2D)$.

Figure 3(b) shows the construction for $D=2$ and $N=4$. Current first advances along the x_1 direction from L_k to the embedded copy of Q_k in L_{k+1}. It then spreads sideways to become uniform on the larger cross-section Q_{k+1}. The final forward step reaches F_N° without further redistribution. The following lemma bounds the energy required for the sideways redistribution.

Lemma 2 (Transport on nested cubes)**.** *For integers $n \ge 1$ and $k \ge 0$, let ρ_k be the uniform probability distribution on $\{-k,\dots,k\}$, extended by zero outside this interval. Let $u_k := \rho_k^{\otimes n}$ be the uniform probability distribution on Q_k. Here u_k denotes a probability distribution, distinct from the scalar proposal probability in* (8)*.*

There is a nearest-neighbor current ψ_k on Q_{k+1} such that

$$\operatorname{div}\psi_k = u_k - u_{k+1},$$

where divergence means outgoing current minus incoming current at each vertex. Thus the current removes the excess mass of u_k and supplies the mass needed to obtain u_{k+1}. Moreover,

$$\sum_{e\subset Q_{k+1}} \psi_k(e)^2 \le \frac{n}{(k+1)^n}. \tag{29}$$

Each unoriented edge is counted once. The sum in (29) *is the energy when every edge has conductance one.*

Proof. We first construct the current in one dimension, then apply this construction one coordinate at a time.

In one dimension, orient every edge to the right. Define the signed current from j to $j+1$ by

$$J_{j+1/2} := \sum_{t=-k-1}^{j} \big(\rho_k(t) - \rho_{k+1}(t)\big), \qquad -k-1 \le j \le k.$$

A negative value means that the current travels to the left. This definition has a simple interpretation: the current crossing an edge equals the total excess mass to its left.

Set the current outside the interval to zero. Subtracting successive cumulative sums gives

$$J_{j+1/2} - J_{j-1/2} = \rho_k(j) - \rho_{k+1}(j).$$

Thus outgoing minus incoming current is exactly the required change in mass at each vertex, including the endpoints.

Direct evaluation gives

$$J_{j+1/2} = -\frac{1}{2k+3} + \frac{2(j+k+1)}{(2k+1)(2k+3)}.$$

These currents increase from $-1/(2k+3)$ to $1/(2k+3)$, so their absolute values are at most $1/(2k+3)$. There are $2k+2$ edges, and hence

$$\sum_{j=-k-1}^{k} J_{j+1/2}^2 \le \frac{2k+2}{(2k+3)^2} \le \frac{1}{k+1}.$$

For general n, expand the distribution one coordinate at a time. After expanding the first r coordinates, the distribution is

$$\nu^{(r)} := \rho_{k+1}^{\otimes r} \otimes \rho_k^{\otimes(n-r)}, \qquad r = 0,\dots,n.$$

In particular, $\nu^{(0)} = u_k$ and $\nu^{(n)} = u_{k+1}$.

At stage r, where $0 \le r < n$, we expand coordinate $r+1$. Fix the other coordinates, denoting their values by $z = (z_i)_{i\ne r+1}$. The points obtained by varying coordinate $r+1$ form a coordinate line. The total mass assigned to this line is

$$w_r(z) := \prod_{i=1}^{r} \rho_{k+1}(z_i) \prod_{i=r+2}^{n} \rho_k(z_i).$$

On this line, use the one-dimensional currents multiplied by $w_r(z)$: the current across its edge from j to $j+1$ is $w_r(z)J_{j+1/2}$.

Doing this on every such line defines a current $\psi^{(r)}$ using only edges in coordinate direction $r+1$. The one-dimensional mass-balance identity shows that

$$\operatorname{div}\psi^{(r)} = \nu^{(r)} - \nu^{(r+1)}.$$

To bound its energy, note that a uniform distribution on m points has sum of squared probabilities equal to $1/m$. Therefore

$$\sum_z w_r(z)^2 = (2k+3)^{-r}(2k+1)^{-(n-r-1)} \le (k+1)^{-(n-1)}.$$

Here empty products are interpreted as 1. Combining this with the one-dimensional energy bound yields

$$\begin{aligned}\sum_e \psi^{(r)}(e)^2 &= \left(\sum_z w_r(z)^2\right)\left(\sum_{j=-k-1}^{k} J_{j+1/2}^2\right)\\ &\le (k+1)^{-(n-1)}(k+1)^{-1}\\ &= (k+1)^{-n}.\end{aligned}$$

Finally, add the currents from all n stages:

$$\psi_k := \sum_{r=0}^{n-1} \psi^{(r)}.$$

Their divergences telescope, giving

$$\operatorname{div}\psi_k = \sum_{r=0}^{n-1} \big(\nu^{(r)} - \nu^{(r+1)}\big) = u_k - u_{k+1}.$$

Each stage uses a different coordinate direction, so no edge is used by two stages. Their energies therefore add:

$$\sum_e \psi_k(e)^2 = \sum_{r=0}^{n-1} \sum_e \psi^{(r)}(e)^2 \le \frac{n}{(k+1)^n}.$$

This also covers $k = 0$. □

We now construct a unit flow whose energy gives the required resistance upper bound. Recall that $\alpha = 1/p > 1$.

Proposition 2 (Resistance upper bound)**.** *Fix $D \ge 1$ and $0 < p < 1$. There exists a finite constant $C'_{D,p}$, independent of N, such that for every integer $N \ge 1$,*

$$R_{\text{eff}}(0 \leftrightarrow F_N^\circ) \le C'_{D,p} \frac{\alpha^N}{N^{D-1}}. \tag{30}$$

Proof. For $D = 1$, the target face is the single point N. Send one unit of current along the path $0, 1, \dots, N$. The edge $k \leftrightarrow k+1$ has conductance $p^{k+1}/2$, so this flow has energy

$$2 \sum_{k=0}^{N-1} \alpha^{k+1} = \frac{2\alpha(\alpha^N - 1)}{\alpha - 1} \le \frac{2\alpha}{\alpha - 1} \alpha^N.$$

Thomson's principle gives the claimed bound.

Now assume $D \ge 2$ and write $n = D - 1$. We construct the flow through the pyramid layers described above. At each layer L_k, the outgoing forward current will be distributed uniformly over its $(2k+1)^n$ vertices.

Forward current. For $k = 0, \dots, N-1$, put current $1/(2k+1)^n$ on each edge from (k, y) to $(k+1, y)$, with $y \in Q_k$. These currents have total value one. Since each edge has conductance $p^{k+1}/(2D)$, their total energy is

$$E_k^{\text{vert}} = (2k+1)^n \frac{(2k+1)^{-2n}}{p^{k+1}/(2D)} = \frac{2D\,\alpha^{k+1}}{(2k+1)^n} \le \frac{2D\,\alpha^{k+1}}{(k+1)^n}.$$

Redistribution within a layer. For $k = 0, \dots, N-2$, the forward current arrives in L_{k+1} uniformly on Q_k, whereas the next forward step requires it to leave uniformly on Q_{k+1}. Use the current ψ_k from Lemma 2 on the lateral edges of L_{k+1} to make this redistribution.

The lemma gives

$$\sum_e \psi_k(e)^2 \le \frac{n}{(k+1)^n}.$$

Each lateral edge in this layer has resistance $2D\alpha^{k+1}$. Consequently, the redistribution energy satisfies

$$E_{k+1}^{\text{hor}} = 2D\alpha^{k+1} \sum_e \psi_k(e)^2 \le \frac{2Dn\,\alpha^{k+1}}{(k+1)^n}.$$

The final forward step, from L_{N-1} into F_N°, requires no further redistribution.

Conservation of current. At a vertex indexed by $y \in Q_j$ in an intermediate layer L_j, where $1 \le j \le N-1$, the incoming forward current is $u_{j-1}(y)$ and the outgoing forward current is $u_j(y)$. Here u_{j-1} is extended by zero outside Q_{j-1}. The lateral current has divergence $u_{j-1}(y) - u_j(y)$, so the total divergence is

$$-u_{j-1}(y) + u_j(y) + \big(u_{j-1}(y) - u_j(y)\big) = 0.$$

The origin supplies one unit of current, and the target face receives one unit, uniformly distributed over its vertices. Thus the construction is a unit flow from 0 to F_N°. For $N = 1$, it consists of just the edge from the origin to the target point. Set the current to zero on every edge outside the construction.

Total energy. Forward edges and lateral edges are disjoint, and different redistribution steps use different layers. Their energies therefore add. By Thomson's principle,

$$\begin{aligned} R_{\text{eff}}(0 \leftrightarrow F_N^\circ) &\le \sum_{k=0}^{N-1} E_k^{\text{vert}} + \sum_{k=0}^{N-2} E_{k+1}^{\text{hor}} \\ &\le 2D(1+n) \sum_{j=1}^{N} \frac{\alpha^j}{j^n}. \end{aligned}$$

It remains to bound this sum by a constant times its final term. Set $a_j := \alpha^j / j^n$. Since

$$\frac{a_{j-1}}{a_j} = \alpha^{-1} \left(\frac{j}{j-1}\right)^n \longrightarrow \alpha^{-1} < 1,$$

there exist $q \in (0,1)$ and an integer $J \ge 1$, depending only on n and p, such that $a_{j-1} \le q a_j$ for every $j \ge J+1$. For $N \ge J$, repeated application of this inequality gives

$$\sum_{j=J}^{N} a_j \le a_N \sum_{\ell=0}^{N-J} q^\ell \le \frac{a_N}{1-q}.$$

Also $a_N \ge a_J$, so

$$\sum_{j=1}^{J-1} a_j \le \frac{\sum_{j=1}^{J-1} a_j}{a_J}\, a_N.$$

Combining these bounds, and enlarging the constant to cover the finitely many cases $N < J$, yields

$$\sum_{j=1}^{N} \frac{\alpha^j}{j^n} \le K_{n,p} \frac{\alpha^N}{N^n}.$$

Since $n = D - 1$, this proves the proposition. □

We can now state and prove the full scaling law.

Theorem 4 (True-chain escape-time scaling)**.** *Fix $D \ge 1$ and $0 < p < 1$, and let $\alpha = 1/p$. For the true Metropolis chain on $\mathbb{Z}^D$ with energy $V(x) = \|x\|_\infty$, started from the origin,*

$$\mathbb{E}_0[\tau_N] = \Theta\left(\frac{\alpha^N}{N^{D-1}}\right) \qquad (N \to \infty). \tag{31}$$

Proof. The lower bound is Theorem 3. For the upper bound, since $\tau_N \le \tau_{F_N^\circ}$, Lemma 1 gives

$$\mathbb{E}_0[\tau_N] \le \mathbb{E}_0[\tau_{F_N^\circ}] \le M_{D,p}\, R_{\text{eff}}(0 \leftrightarrow F_N^\circ) \le M_{D,p}\, C'_{D,p} \frac{\alpha^N}{N^{D-1}},$$

using Proposition 2. Since $M_{D,p} < \infty$ depends only on D and p, combining with the lower bound yields (31). □

Remark 5. *The shell-averaged chain identifies the correct bottleneck scale: the resistance across shell k is of order α^k/k^{D-1}. The lower bound shows this is unavoidable (collapsing shells only helps), while the explicit face flow shows that the true chain can exploit the full $(D-1)$-dimensional surface area of each shell, so no larger asymptotic obstruction is present.*

VI. Numerical Results

A. Numerical Method

We compare the true Metropolis mean escape time $H_D(N)$ with the shell-quotient prediction $Q_D(N)$ for $D = 1, 2, 3$. Both count attempted moves, including rejected proposals. We compute $H_D(N)$ by grouping states that differ only in coordinate signs or order and solving the resulting sparse linear systems for escape probabilities and excursion times. This symmetry reduction preserves the true dynamics exactly. We evaluate $Q_D(N)$ using the closed-form expression in Theorem 1.

B. Dimensional Speedup

Figure 4(a) shows that higher dimension reduces escape time in the basins tested at $p = 0.55$. The separation grows with depth: at $N = 25$, escape is about 6.6 times faster in two dimensions and 28.4 times faster in three dimensions than in one dimension. This illustrates the polynomial dimensional advantage in the proved scaling $H_D(N) = \Theta(p^{-N}/N^{D-1})$.

Figure 4(b) shows how the speedup depends on the acceptance probability at $N = 20$. The three-dimensional walk escapes about 243 times faster than the one-dimensional walk at $p = 0.15$, but only about 1.14 times faster at $p = 0.9$. The dimensional advantage is therefore much larger at strong uphill rejection in this example. For shallow basins, the ordering can reverse: at $N = 2$ and $p = 0.9$, $H_1 \approx 4.57$ whereas $H_2 \approx 5.02$.

C. Accuracy of the Shell Quotient

The upper panels of Figure 5 compare $Q_D(N)$ with $H_D(N)$. Their curves largely overlap on the logarithmic scale, so the lower panels show the signed relative discrepancy

$$\delta_D(N,p) := 100\left(\frac{Q_D(N)}{H_D(N)} - 1\right).$$

A negative value means that the quotient underestimates the true mean escape time. In one dimension, $Q_1 = H_1$ exactly.

At $N = 25$ and $p = 0.2$, the quotient underestimates the true escape time by approximately 1.4% in two dimensions and 2.8% in three dimensions. At $p = 0.9$, the corresponding discrepancies are larger: 14.5% and 25.5%. At low p, the discrepancy decreases over the larger depths shown, whereas at $p = 0.9$ it changes little at those depths. The quotient captures the escape-time scale, while its quantitative accuracy depends on dimension, acceptance probability, and basin depth.

The discrepancy persists even without uphill rejection: at $p = 1$ and $N = 40$, numerical calculations give $H_2/N^2 \approx 1.1786$ and $H_3/N^2 \approx 1.3488$, corresponding to quotient discrepancies of approximately -15.15% and -25.86%, respectively. These values provide context for the nearly flat discrepancy curves at $p = 0.9$, without establishing a common asymptotic limit.

VII. Conclusion

In this paper, we have extended the finite-time analysis of constant-temperature simulated annealing from one dimension to D-dimensional cubic basins defined by the infinity norm. We derived a closed-form expression for the mean escape time of the shell-averaged Metropolis chain (Theorem 1), proved that its asymptotic scaling is $\Theta(\alpha^N/N^{D-1})$ (Theorem 2), and established that the true Metropolis chain, without shell averaging, has the same scaling (Theorem 4), via a lower bound from Rayleigh monotonicity and a matching upper bound from an explicit Thomson flow.

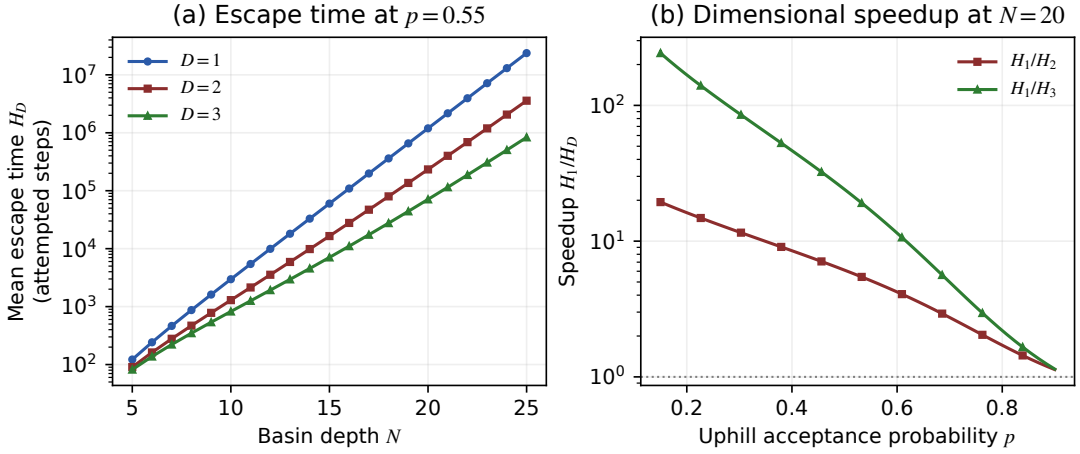


Fig. 4. Dimensional dependence of the true Metropolis escape time. (a) Mean escape time versus basin depth at $p = 0.55$, for $D = 1, 2, 3$ and integer depths $N = 5, \ldots, 25$. (b) Speedup relative to one dimension, H_1/H_D, at $N = 20$ for 50 acceptance probabilities between 0.15 and 0.9. Values above one indicate faster escape in dimension D. Both vertical axes are logarithmic.

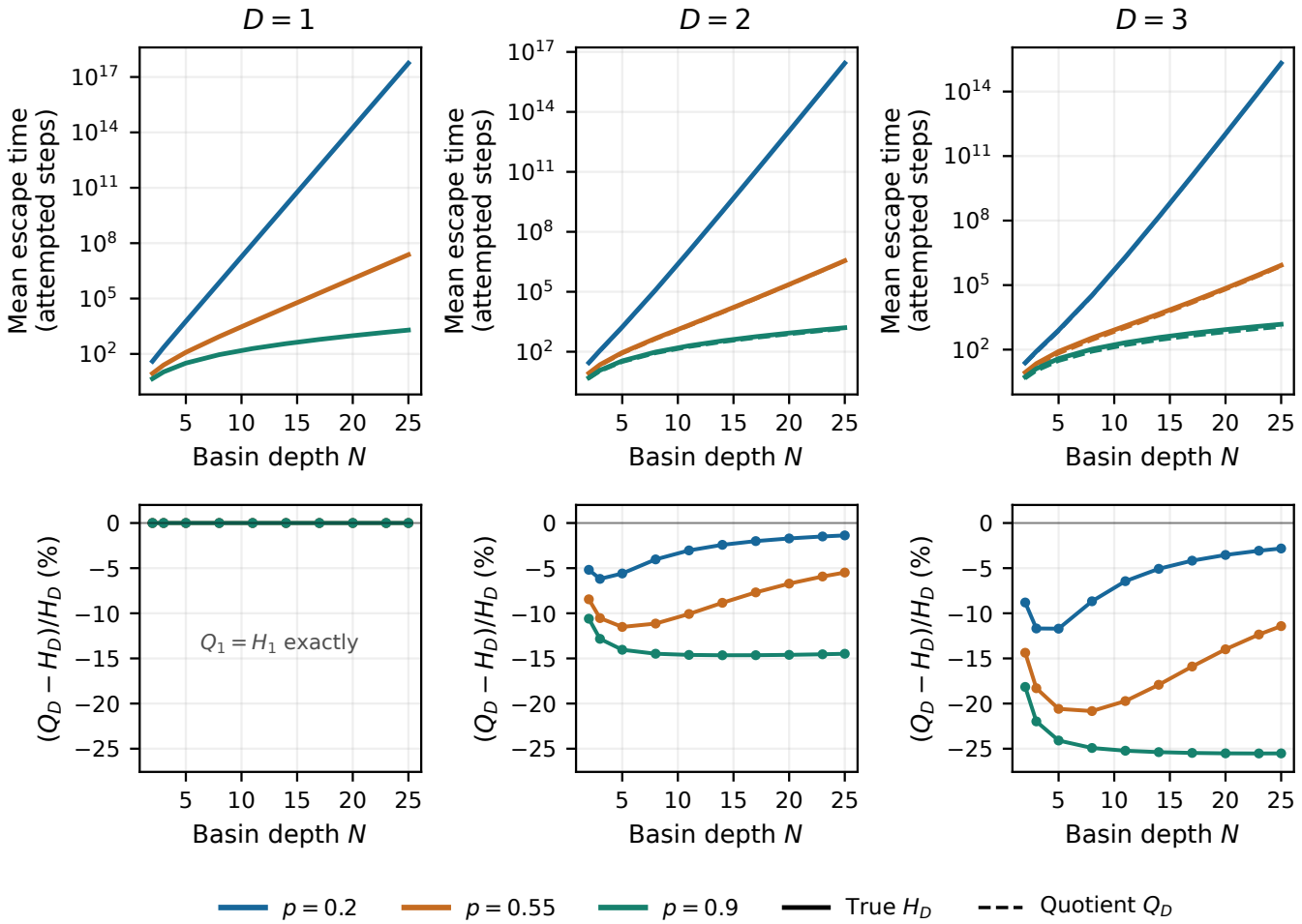


Fig. 5. True Metropolis mean escape time $H_D(N)$ (solid) and shell-quotient mean $Q_D(N)$ (dashed) versus basin depth (top), with signed relative discrepancy $100(Q_D/H_D - 1)$ in percent (bottom). Columns correspond to $D = 1, 2, 3$; colors indicate $p = 0.2, 0.55, 0.9$. The computed depths are $N = 2, 3, 5, 8, 11, 14, 17, 20, 23, 25$. Times count attempted moves, and the upper panels use logarithmic vertical axes. For $D = 1$, the means coincide exactly and only the solid curves are shown.

The central conclusion is specific to deep cubic basins: for fixed D and $0 < p < 1$, the mean time to first reach the outer shell (any site with $\|x\|_\infty = N$) is $\Theta(p^{-N}/N^{D-1})$ attempted steps. The polynomial factor reflects the available boundary area and yields an asymptotic dimensional speedup within this family.

Several directions for future work remain. First, extending the analysis to non-cubic basins (e.g., basins defined by the ℓ^1 or ℓ^2 norm) would test the robustness of the polynomial speedup to basin geometry. Second, the multi-basin analysis of [1] could be extended to investigate escape and global hitting times in multidimensional multi-basin landscapes. Third, studying continuous-state counterparts could test whether the discretization effects reported in [1] persist in higher dimensions. Fourth, determining a leading constant for the true-chain escape time would sharpen Theorem 4; the constant in Remark 3 currently applies only to the quotient chain.

Code and expanded proofs.

The numerical notebook and expanded proofs in Lamport's hierarchical proof format are available at https://github.com/ANRGUSC/metropolis-highD/. The notebook reproduces the numerical results and figures, while the supplement makes intermediate claims and dependencies explicit to assist with proof verification. These structured proofs are not machine-checked formal proofs.

AI Use Statement

The authors made substantial use of ChatGPT, Claude, and Grok in developing the mathematical results, numerical computations, illustrations, exposition, and formatting of this paper. The authors accept full responsibility for the paper's contents.

References

[1] H. Ramachandran and B. Krishnamachari, "Escaping local minima: A finite-time Markov chain analysis of constant-temperature simulated

annealing," in *Proc. IEEE SoutheastCon*, 2026, pp. 1–6. doi: 10.1109/SoutheastCon63549.2026.11476393. Preprint: arXiv:2602.09398.

[2] S. Geman and D. Geman, "Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 6, pp. 721–741, 1984.

[3] B. Hajek, "Cooling schedules for optimal annealing," *Mathematics of Operations Research*, vol. 13, no. 2, pp. 311–329, 1988.

[4] O. Catoni, "Rough large deviation estimates for simulated annealing: Application to exponential schedules," *The Annals of Probability*, vol. 20, no. 3, pp. 1109–1146, 1992.

[5] M. P. Desai and V. B. Rao, "Finite-time behavior of slowly cooled annealing chains," *Probability in the Engineering and Informational Sciences*, vol. 11, no. 2, pp. 137–176, 1997.

[6] R. Holley, S. Kusuoka, and D. Stroock, "Asymptotics of the spectral gap with applications to the theory of simulated annealing," *Journal of Functional Analysis*, vol. 83, no. 2, pp. 333–347, 1989.

[7] D. Mitra, F. Romeo, and A. Sangiovanni-Vincentelli, "Convergence and finite-time behavior of simulated annealing," *Advances in Applied Probability*, vol. 18, no. 3, pp. 747–771, 1986.

[8] V. V. Palyulin and R. Metzler, "How a finite potential barrier decreases the mean first-passage time," *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2012, no. 03, p. L03001, 2012.

[9] G. O. Roberts, A. Gelman, and W. R. Gilks, "Weak convergence and optimal scaling of random walk Metropolis algorithms," *The Annals of Applied Probability*, vol. 7, no. 1, pp. 110–120, 1997.

[10] A. Molavi, A. Xu, E. Cecchetti, S. Tannu, and A. Albarghouthi, "Generating compilers for qubit mapping and routing," *arXiv preprint arXiv:2508.10781v2*, 2025.

[11] C. Sechen and A. L. Sangiovanni-Vincentelli, "The TimberWolf placement and routing package," *IEEE Journal of Solid-State Circuits*, vol. 20, no. 2, pp. 510–522, 1985.

[12] S. Kirkpatrick, C. D. Gelatt Jr., and M. P. Vecchi, "Optimization by simulated annealing," *Science*, vol. 220, no. 4598, pp. 671–680, 1983.

[13] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, and E. Teller, "Equation of state calculations by fast computing machines," *The Journal of Chemical Physics*, vol. 21, no. 6, pp. 1087–1092, 1953.

[14] P. G. Doyle and J. L. Snell, *Random Walks and Electric Networks*, Carus Mathematical Monographs, vol. 22. Mathematical Association of America, 1984.

[15] D. A. Levin and Y. Peres, *Markov Chains and Mixing Times*, 2nd ed. American Mathematical Society, 2017.

[16] P. Tetali, "Random walks and the effective resistance of networks," *Journal of Theoretical Probability*, vol. 4, no. 1, pp. 101–109, 1991.

[17] Y. Dauphin, R. Pascanu, C. Gulcehre, K. Cho, S. Ganguli, and Y. Bengio, "Identifying and attacking the saddle point problem in high-dimensional non-convex optimization," in *Advances in Neural Information Processing Systems 27 (NeurIPS)*, 2014.

[18] M. Bellitti, F. Ricci-Tersenghi, and A. Scardicchio, "Entropic barriers as a reason for hardness in both classical and quantum algorithms," *Physical Review Research*, vol. 3, p. 043015, 2021.

[19] D. J. Bicout and A. Szabo, "Entropic barriers, transition states, funnels, and exponential protein folding kinetics: A simple model," *Protein Science*, vol. 9, no. 3, pp. 452–465, 2000.

[20] A. Bovier, M. Eckhoff, V. Gayrard, and M. Klein, "Metastability and low lying spectra in reversible Markov chains," *Communications in Mathematical Physics*, vol. 228, pp. 219–255, 2002.

[21] A. Bovier and F. den Hollander, *Metastability: A Potential-Theoretic Approach*, Grundlehren der mathematischen Wissenschaften, vol. 351. Springer, 2015.

[22] J. G. Kemeny and J. L. Snell, *Finite Markov Chains*. Springer-Verlag, 1976, Sec. 6.3.